

Evidence for Three Subpopulations of Merging Binary Black Holes at Different Primary Masses

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With the release of the fourth LIGO-Virgo-KAGRA gravitational-wave catalog (GWTC-4), we are starting to gain a detailed view of the population of merging binary black holes. The formation channels of these black holes is not clearly understood, but different formation mechanisms may lead to subpopulations with different properties visible in gravitational-wave data. Adopting a phenomenological approach, we find GWTC-4.0 data supports the presence of at least three subpopulations, each associated with a different range of black hole mass and with sharp transition boundaries between them. Each subpopulation is characterized by different distributions for either the mass ratios, the black-hole spin magnitudes or both. Subpopulation $\mathcal{A}$ with primary mass $m_1 \leq 27.7^{+4.1}_{-3.4} M_\odot$ (90% credibility), is characterized by a nearly flat mass ratio distribution $q = m_2/m_1$, and by small spin magnitudes ($\chi \leq 0.5^{+0.1}_{-0.1}$). Subpopulation $\mathcal{B}$, with $27.7^{+4.1}_{-3.4} M_\odot \leq m_1 \leq 40.2^{+4.7}_{-3.2} M_\odot$, has a much sharper preference for mass ratio $q \approx 1$. Subpopulation $\mathcal{C}$, with $m_1 \geq 40.2^{+4.7}_{-3.2} M_\odot$, has support for large spin magnitudes, and tentative support for mass ratios $q \approx 0.5$. We interpret these transitions as evidence for multiple subpopulations, each potentially associated with a different formation pathways. We suggest putative formation scenarios for each subpopulation, and explore chemically homogeneous evolution, population III stars and dynamical formation channels as an explanation for subpopulation $\mathcal{B}$. Our findings for subpopulation $\mathcal{C}$ are largely consistent with recent claims of hierarchical mergers but with some curious differences in properties.

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Introduction—The sensitivity of gravitational wave (GW) detectors has increased massively in the last ten years [1,2] as a result of technological upgrades [3–6] and an improved understanding of instrumental noise [7]. With the recent GWTC-4.0 catalog [8–10] by LIGO-Virgo-KAGRA (LVK) [11–13], the number of confidently detected binary black hole (BBH) mergers has now surpassed 150, offering an unprecedented view into the nature and formation of these systems and their populations [14]. In the early days of GW astronomy, it seemed plausible that just a small number of detections might enable us to determine the primary formation channel of BBH mergers (e.g., [15–17]). However, this hope has since dissipated as GW data has unveiled a rich and complex picture of the population of merging binaries, not easily explained with a single formation channel. Furthermore, uncertainties in the physics underlying black hole formation and merger among the various possible channels make it hard to obtain robust predictions (see Ref. [18] for a review). The relative contributions of various BBH formation channels is still a topic of active study (e.g., [19–22]).

Several analyses have claimed evidence of BBH subpopulations. References [14–16,23–27] used BBH spin tilts

to discriminate between different subpopulations. While promising from a theoretical perspective [28–31], using this as a clean discriminator has proven difficult. First, precise measurements of BBH spin tilts with GWs is difficult. Second, recent studies have challenged commonly held assumptions and shown that these models can be subject to significant theoretical uncertainties [32–35]. There is also evidence for global correlations between various parameters [14,36–39] but interpreting them astrophysically is not straightforward.

Another potentially fruitful way is to adopt a phenomenological approach: to look for subpopulations that have clear and distinguishable properties. If two groups of binaries are found to have different distributions for some parameters like mass ratio, spin or redshift, one can then infer that the relevant physical processes behind their formations are different [40]. Several studies have pursued this strategy to search for subpopulations with interesting properties [14,27,41–51].

It can be particularly informative to find subpopulations separated by clear transition masses. Knowing both the BBH mass scales and their properties can enable us to constrain and infer their formation physics. This is the approach we take in this Letter by searching for subpopulations with distinct primary black hole masses. Using cumulative GWTC-4.0 data [10], we find evidence for at

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least three such subpopulations, each characterized by different mass ratio pairing or different spin distributions. Crucially, the data strongly prefers the interpretation that all of these subpopulations exist in conjunction.

Methods and results—We model the BBH population with simple phenomenological components. We use the strongly parametrized fiducial models from Ref. [14]. The primary mass distribution $p(m_1)$ is a broken power law with two Gaussian peaks. The spin magnitude distributions $p(\chi)$ are (independently and identically distributed) truncated normals, and the spin tilt distributions $p(\cos\theta)$ are mixtures between an isotropic component, and an aligned component modeled with a truncated Gaussian distribution [25,52]. The redshift distributions $p(z)$ are modeled as a power law in $(1+z)$ [53]. Finally, we model the mass-ratio distributions, $p(q)$, where $q = m_2/m_1$, either as a power law or a truncated normal.

A key feature of our work is that the distributions of z , q , χ and $\cos\theta$ are allowed to change shape at up to three transition masses, corresponding to four subpopulations [54]. The functional form of the distribution may be the same on either side of the transition, but the hyperparameters controlling the shape are allowed to have distinct values. For example, our model can accommodate a scenario where binaries with $m_1 < 30M_\odot$ are characterized by a mass ratio distribution of $p(q) \propto q^2$, but binaries with $m_1 > 30$ are characterized by a mass ratio distribution $p(q) = \text{const.}$ More details about model parameterization and the priors used can be found in Supplemental Material [55].

We use standard hierarchical Bayesian inference techniques [56–58] for inferring the BBH population properties. We analyze data cumulative to GWTC-4.0 [10,59,60,70,71] including all BBHs mergers with a false alarm rate less than 1 yr^{-1} . We do not include GW190814 [72] because of the ambiguous nature of the secondary. We also do not include GW231123 [73] in our analysis, driven by concerns that the seemingly extremal spins makes it susceptible to poorly understood waveform systematics. Moreover, if its measured properties are real, the most parsimonious explanation would likely put it in a class of its own highly spinning, ultramassive BBH systems [74].

We then tested several models allowing up to three transition masses. These yield a more-or-less consistent picture of the BBH population, wherein we find subpopulations with different mass-ratio or spin magnitude distributions, but no statistically significant evidence that the distribution of other parameters have a transition. We relied on Bayes factors for model selection and to avoid over fitting the data. Bayes factors carry an Occam penalty for more complex models and ensure that a more flexible model is selected only when its flexibility is required by the data. We direct the reader to Sec. 3 of Ref. [57] for an overview on Bayes factors and Bayesian model selection in gravitational-wave astronomy.

The rest of this Letter focuses on the model with the highest Bayes factor. All quoted uncertainties correspond to 90% credible levels unless stated otherwise.

We find that the data strongly prefers three distinct subpopulations with two transition masses. The three subpopulations differ in either their mass-ratio distributions, their spin magnitude distributions or both. For brevity we refer to this model as MULTIPOP. An overview of our main results with MULTIPOP is shown in Fig. 1. Formally, the mass-ratio distribution of MULTIPOP is modeled as

$$p(q) \propto \begin{cases} q^{\beta_0}, & \text{if } m_1 \leq m'_{AB}, \\ q^{\beta_1}, & \text{if } m'_{AB} \leq m_1 \leq m'_{BC}, \\ \mathcal{N}(q|\mu_q, \sigma_q), & \text{if } m_1 > m'_{BC}, \end{cases} \quad (1)$$

and the spin-magnitude distribution as

$$p(\chi) \propto \begin{cases} \mathcal{N}(\chi|\mu_{\chi_0}, \sigma_{\chi_0}), & \text{if } m_1 \leq m'_{BC}, \\ \mathcal{N}(\chi|\mu_{\chi_1}, \sigma_{\chi_1}), & \text{if } m_1 > m'_{BC}. \end{cases} \quad (2)$$

Redshift and spin tilts are found to need just a single global distribution.

We measure the transition masses separating these subpopulations at $m'_{AB} = 27.7^{+4.1}_{-3.4} M_\odot$ and $m'_{BC} = 40.2^{+4.7}_{-3.2} M_\odot$. These are statistically consistent with transition values individually inferred elsewhere in the literature. In particular, Ref. [41] finds a transition in $p(q)$ at $29.3^{+5.7}_{-4.0} M_\odot$ with BBHs above this mass have a stronger preference for equal-mass pairing [76]. Meanwhile Ref. [27] and Ref. [46] measure a transition in spins at $m_1 = 41.56^{+20.64}_{-8.55}$ (1σ credible interval) and $m_1 = 46^{+7}_{-5} M_\odot$, respectively. More recently, Ref. [48] also found a transition at $45^{+5}_{-4} M_\odot$ in the secondary mass distribution. Since this Letter was made public, newer work in the literature also found evidence to support such transitions; in particular Refs. [77,78] find a transition at $\sim 45 M_\odot$ to a population of binaries with more rapid spins, while Ref. [79] finds evidence for a gentler transition in the secondary mass that is still consistent with our result.

The MULTIPOP model is preferred over the fiducial strongly parametrized model from Ref. [14] (which we refer to as GWTC4-FIDUCIAL) by a natural log Bayes factor of 12.3. We direct the reader to the Supplemental Material [55] and to Ref. [14] for more details about GWTC4-FIDUCIAL. We now describe the characteristic properties of each of the subpopulations. (i) *Subpopulation A*: $m_1 \leq 27.7^{+4.1}_{-3.4} M_\odot$ —The mass ratio distribution is somewhat flat; we infer $\beta_0 = -0.8^{+2.4}_{-1.9}$. The spin-magnitude distribution is mostly localized within $\chi \leq 0.5^{+0.1}_{-0.1}$ —see Fig. 2. Since this subpopulation includes a large range of black hole masses from $3\text{--}28 M_\odot$, we speculate that it may actually include contributions from multiple formation channels. (ii) *Subpopulation B*: $27.7^{+4.1}_{-3.4} M_\odot < m_1 \leq 40.2^{+4.7}_{-3.2} M_\odot$ —This subpopulation is characterized by a stronger preference for equal mass

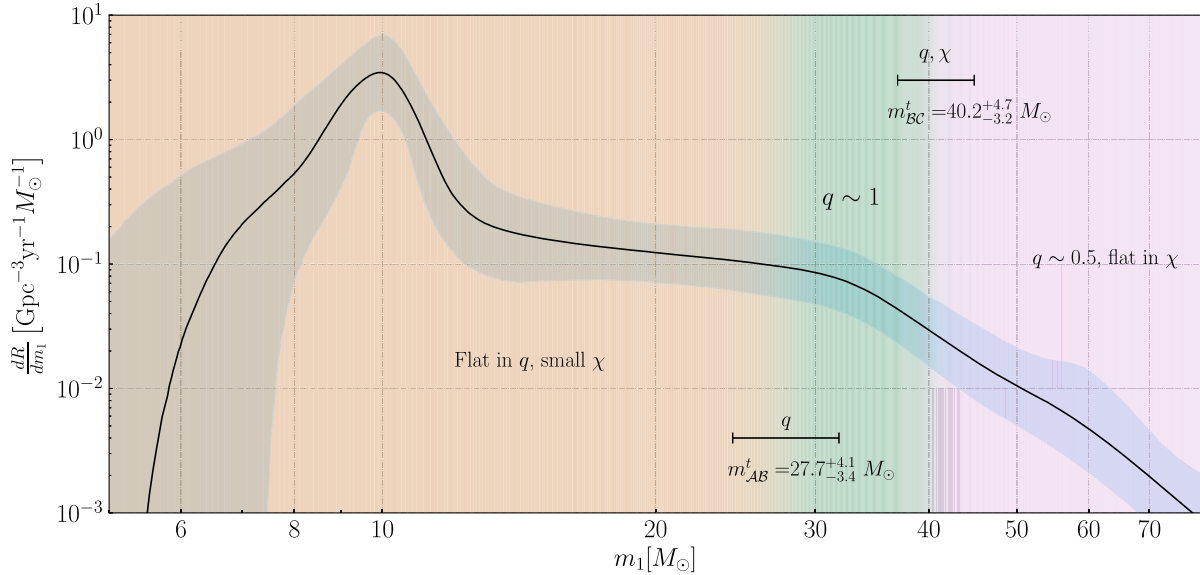


FIG. 1. The three subpopulations inferred with the MULTIPOP model are shown as three colored regions along the primary mass spectrum. The transition regions between the colors correspond to the inferred transition masses. Subpopulation $\mathcal{A}$, characterized by $m_1 \lesssim 28M_\odot$ has an almost flat mass ratio distribution and a $\chi \lesssim 0.5$. Subpopulation $\mathcal{B}$, characterized by $28M_\odot \lesssim m_1 \lesssim 40M_\odot$ has a mass-ratio distribution that is more sharply peaked at $q = 1$ while retaining the same spin distribution as subpopulation $\mathcal{A}$. Subpopulation $\mathcal{C}$, at $m_1 \gtrsim 40M_\odot$ shows support for a spin-magnitude distribution with significant support at $\chi > 0.5$ and tentative preference for $q \sim 0.5$. The solid black curve and the shaded region around it correspond to the median population predictive distribution and the 90% credible levels of the primary mass mass spectrum.

pairing; we infer $\beta_1 = 3.6^{+3.0}_{-2.2}$ and $\beta_1 > \beta_0$ at 97% credibility. The spin properties of this subpopulation are indistinguishable from those of subpopulation $\mathcal{A}$. Interestingly, by allowing for these two transitions, evidence for a peak in $p(m_1)$ at $m_1 \approx 35M_\odot$ is diminished. However, we see tentative signs of an peak in $p(m_2)$ at a slightly lower mass; see Fig. 3. (iii) *Subpopulation C*: $m_1 > 40.2^{+4.7}_{-3.2}M_\odot$ —The subpopulation is characterized by a wider spin-magnitude distribution that is incompatible with the first two; see Fig. 2. The mass-ratio distribution also changes. We find tentative evidence that $q \approx 0.5$ for this subpopulation, see Figs. 4 and 5. But this is not conclusive and more data are required to confirm this possibility.

We now turn to measuring the statistical significance of individual features and transitions. We define two different metrics for this purpose. The first is the relative Bayes factor, $\mathcal{B}_R$, which is the gain in evidence when a new feature is added to GWTC4-FIDUCIAL model. For example, we can compare the evidence after adding a transition in $p(q)$ to GWTC4-FIDUCIAL with the evidence without any transitions. We also define the differential Bayes factor, $\mathcal{B}_D$ which compares the evidence of the MULTIPOP model [Eqs. (1) and (2)] to the evidence when a feature is taken out. We direct the reader to Eqs. (S7) and (S8) in Supplemental Material [55] for formal definitions. These two metrics allow us to gauge the significance of a feature both on its own, and in conjunction with other features that capture the presence of multiple subpopulations. Table I shows the $\ln \mathcal{B}_R$ and $\ln \mathcal{B}_D$ values for the features identified by the MULTIPOP model.

The individual feature most strongly supported by data is that subpopulation $\mathcal{C}$ has a distinct spin-magnitude distribution with support for $\chi > 0.5$. When we allow the mass ratio distribution to simultaneously change for this subpopulation, this significantly bolsters evidence for a transition at $m_1 \geq m_{BC}^t$. The hypothesis that there are three subpopulations with distinct mass-ratio distributions (when keeping the spin magnitude distribution fixed) is strongly supported with $\ln \mathcal{B}_R$ and $\ln \mathcal{B}_D$ of 6.0 and 4.5, respectively. Finally, we note that when we tested a variant of MULTIPOP that models the mass-ratio distribution of subpopulation $\mathcal{C}$ as a power law, it is disfavored with a $\log \mathcal{B}$ of 4.8 compared to the standard version described in Eq. (1).

Astrophysical interpretation—We turn now to the astrophysical interpretation of these subpopulations, guided not just by the observed transitions in $p(q)$ and $p(\chi)$ but also the lack of any transitions in spin-tilt or redshift distributions.

Subpopulation A: This subpopulation is characterized by a somewhat flat mass-ratio distribution and spin magnitudes constrained at $\chi \leq 0.5$. Astrophysical observations of low mass stars find highly effective angular momentum transport [80], which extrapolated to more massive stars imply black hole spins with $\chi \lesssim 0.01$ [81,82]. The fact that we see $\chi > 0.01$ for this subpopulation implies that they have been spun up or retained their angular momentum in some other way. Alternatively, such assumptions of extremely-efficient angular momentum transport might be incorrect for massive stars; models based on the Taylor-dynamo [83] suggest

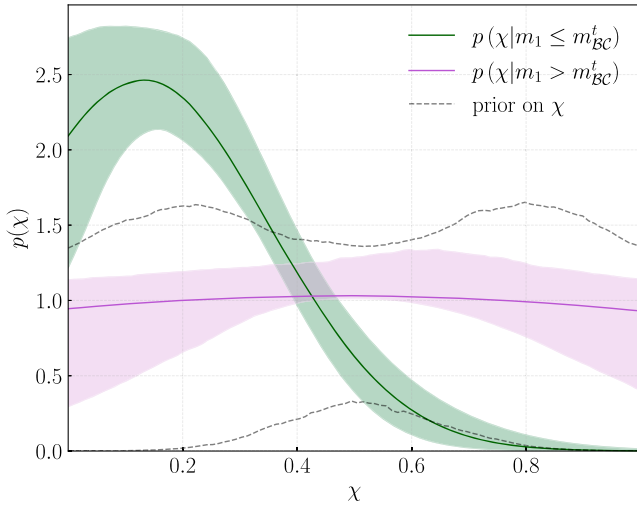


FIG. 2. The spin magnitude distributions for different subpopulations of binary black holes. In this and all subsequent figures, the solid curves are the population predictive distributions and the shaded regions are the 90% credible interval. The 90% credible range of the population prior on the spin magnitudes (see Table S1 in the Supplemental Material [55]) is shown with the dashed lines. Subpopulation $\mathcal{A}$ and subpopulation $\mathcal{B}$ are both characterized by comparatively small spins (green) with $\chi < 0.5^{+0.1}_{-0.1}$. On the other hand, subpopulation $\mathcal{C}$ (magenta) is characterized by a comparatively broader distribution with significant support for $\chi > 0.5$.

that the spin of isolated black holes can be as high as $\chi \approx 0.1$ [84–86].

It is also possible that multiple formation channels contribute to the subpopulation below $m_1 \leq m_{AB}^t$, especially

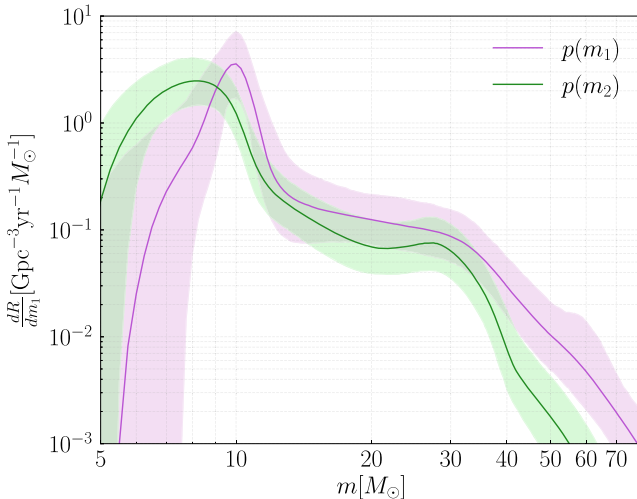


FIG. 3. The reconstructed marginalized distributions of primary mass, m_1 and secondary mass, m_2 . The distribution of m_2 is characterized by a bump at $30M_\odot$, which is not immediately apparent in the distribution of m_1 . Both distributions fall above $40M_\odot$, but the m_2 distribution falls more rapidly than the m_1 distribution.

around the $10M_\odot$ peak which is the global maxima of the mass BBH mass spectrum [14,87,88]. There is tentative observational evidence that this is the case with some analyses finding substructure among the subpopulation that makes up the peak [14,42,47]. Our exploratory studies, when allowing for a transition around the $10M_\odot$ peak find broad posteriors for transition masses, perhaps indicating some substructure that is not well fit by our phenomenological models. Nevertheless, we find no clear evidence for multiple subpopulations there. We touch on this point again at the end of this Letter.

Subpopulation $\mathcal{B}$: This subpopulation is characterized by a stronger preference for equal mass pairing. The preference for a bump at $\sim 35M_\odot$ in $p(m_1)$ —first noted in Ref. [89]—disappears with the more complicated pairing distribution of MULTIPOP; see Figs. 1 and 3. We find that the absence of an m_1 bump is a consequence of subpopulation $\mathcal{B}$ having a distinct mass ratio distribution; when forced to share a common mass ratio distribution with either subpopulation $\mathcal{A}$ or subpopulation $\mathcal{C}$, the peak reappears. However, we find hints of a bump in $p(m_2)$ at $\sim 30M_\odot$; see Fig. 3. A similar result, albeit using the GWTC-3 catalog, was reported by Ref. [90] with a model that independently models and pairs $p(m_1)$ and $p(m_2)$. Thus, we propose that the previously identified m_1 bump might be due to model misspecification [91]. Future data and analysis with different models will help to shed more light on the possibility of misspecification and the true nature of this feature.

Recently, Ref. [92] conducted an investigation of BBHs at the $35M_\odot$ bump. In effect; they studied binaries equivalent to our subpopulation $\mathcal{B}$. Comparing GW inferences to a suite of different population-synthesis results, they conclude that none of the formation channels considered are consistent with the observed features of subpopulation $\mathcal{B}$.

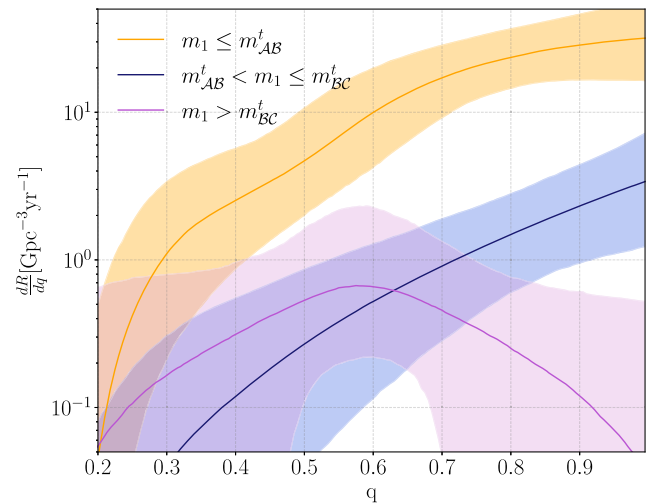


FIG. 4. The reconstructed merger rate of the three subpopulations as a function of the mass ratio. The distribution for subpopulation $\mathcal{C}$ exhibits a tentative peak at $q \approx 0.5$.

TABLE I. Relative Bayes factor $\mathcal{B}_R$ and differential Bayes factors $\mathcal{B}_D$ for the features that define the MULTIPOP model. The relative Bayes factor measures the gain in evidence when a feature is added on top of the GWTC4-FIDUCIAL. The differential Bayes factor compares the evidence of the MULTIPOP model with and without a feature. The transition points, m_{AB}^t and m_{BC}^t (see Fig. 1), are localized at $27.7^{+4.1}_{-3.4} M_\odot$ and $40.2^{+4.7}_{-3.2} M_\odot$, respectively.

Feature or model	$\ln \mathcal{B}_R$	$\ln \mathcal{B}_D$
MULTIPOP model	12.3	
$p(\chi)$ transition at $m_1 = m_{BC}^t$	7.9	6.3
$p(q)$ and $p(\chi)$ both transition at $m_1 = m_{BC}^t$	10.1	11.5
$p(q)$ transitions at both $m_1 = m_{AB}^t$ and m_{BC}^t	6.0	4.5
$p(q)$ transition at $m_1 = m_{BC}^t$	2.7	4.0
$p(q)$ transition at $m_1 = m_{AB}^t$	0.8	2.2

While we mostly agree with their assessment, we highlight two potentially promising formation scenarios. The first is chemically homogeneous evolution (CHE) [93–97]. Unlike Ref. [92], we find a gentler and flatter $p(m_1)$, but still with a preference for equal-mass pairing, consistent with CHE [98]. Perhaps a more difficult thing to reconcile with this formation scenario is that we do not find any difference in spin magnitude or tilt distributions between subpopulation $\mathcal{B}$ and subpopulation $\mathcal{A}$. While CHE typically involves stellar progenitors that are rotating at a significant fraction of their critical speed, the spins of the subsequent black holes are much more uncertain, with predictions ranging from $\chi \approx 0.4$ – 1 [97,99,100]. Subpopulation $\mathcal{B}$ might be consistent with CHE formation if the black holes are spun up to $\chi \approx 0.4$; see Fig. 2.

Another potential origin for this subpopulation is from population III stars. Multiple studies robustly identify a characteristic excess of BBH mergers with masses around $30 M_\odot$ [101–104] from population III stars. However, the star-formation rate (and hence the merger rate) of such systems should naturally peak at high redshift. While some fraction can experience long delay times and merge at low redshift, a degree of fine tuning is likely required to ensure the same star-formation history as the rest of the BBHs subpopulation. However, we caution that this conclusion can be susceptible to model misspecification as we only looked for (and failed to find) different redshift distributions at clearly separated m_1 ranges.

Some studies have suggested a dynamical origin for this subpopulation [44]. Canonically, such a population should have randomly aligned spins [29] and—assuming highly efficient angular momentum transport [82]—very small spin magnitudes. As we do not find evidence for a separate spin-magnitude or tilt distribution from Subpopulation $\mathcal{A}$, this interpretation is somewhat at odds with our results. However, the caveat about assumptions of extremely efficient angular momentum transport for massive stars also apply here and, as previously mentioned, models based on the Taylor-dynamo [83] suggest that the spin of isolated

black holes can be as high as $\chi \approx 0.1$ [84–86]. Furthermore, recent suggestions of stellar collisions in dense clusters [35] might be able to produce aligned spinning BBHs through dynamical scenarios.

Finally, it is unlikely that this subpopulation originates from pair-instability physics, a possibility that attracted much attention in the early days of GW astronomy [105,106]. However, multiple studies recently have argued that the putative pileup due to pair-instability physics should lie at least above $40 M_\odot$ (for e.g. [107–111]), and more likely above $45 m_\odot$ —beyond the upper limit of this subpopulation. Although, for an alternative point of view see Refs. [112,113]. In any case, with the more complex pairing model introduced in this work, evidence of a pileup at $35 M_\odot$ is diminished. We find tentative support for a bump in m_2 at $\approx 30 M_\odot$, but this may be too low of a scale for pair instability physics to be relevant.

Subpopulation $\mathcal{C}$: The idea that massive black holes might be formed from hierarchical mergers of earlier generations of black holes has been explored with some of the first unexpectedly massive BBHs detected by the LVK (for, e.g., [114–120]). Recently, this hypothesis has gained traction because of the discovery of an apparent transition in the effective spin distribution [27,45,46] and an apparent gap in the distribution of secondary mass [48].

Our inference of the mass-ratio distribution subpopulation $\mathcal{C}$ —with a mean at $q \approx 0.5$ —provides further tentative support for the hierarchical interpretation as this qualitatively resembles the distribution one might expect from the merger of a second-generation and a first-generation black holes. While the inferred spin distribution is very broad, it does have support at $\chi \sim 0.7$ (Fig. 2), consistent with the interpretations of Refs. [27,45,46,48].

On the other hand, this distribution also has support at small χ , indicating that a fraction of this subpopulation has small spins. Furthermore, we found that a variant of MULTIPOP that allowed only $p(\chi_1)$ to transition at m_{BC}^t is disfavored by $\ln \mathcal{B} = 1.5$ compared to the version that lets both $p(\chi_1)$ and $p(\chi_2)$ transition. Another potential inconsistency is that we find only a transition in the distribution of spin magnitudes but not of spin tilts between subpopulation $\mathcal{C}$ and the rest of BBH population. These features, if confirmed, might suggest that hierarchical mergers in dense star clusters cannot fully explain subpopulation $\mathcal{C}$. Alternate scenarios such as CHE [97,99,100] and accretion-based spin up in AGN disks [121] might better explain some of these features. However, it is hard to see how either of these explanations would explain the preference for $q \approx 0.5$ if that is confirmed by future data. Conversely, alternative accretion scenarios have been proposed that might be able to explain these properties better [122,123].

A few other analyses in the literature also seem to suggest that in rapidly spinning subpopulations, both black holes possess rapid spins [43,75]. Recently, Ref. [47] also inferred a similar result when modeling component spins—finding

that both χ_1 and χ_2 have support at high spin magnitudes for the high mass subpopulation (see their Fig. 6). However, they also infer a different spin tilt distribution, perhaps suggesting that any inference about this subpopulation can be particularly sensitive to model specification and misspecification.

Conclusion—In this Letter, we investigate the possibility of multiple subpopulations of BBHs each associated with a different range of black hole masses. We find strong evidence that there are at least three such subpopulations with different mass-ratio and spin-magnitude distributions, with transitions between them inferred at $27.7^{+4.1}_{-3.4}M_\odot$ and $40.2^{+4.7}_{-3.2}M_\odot$. A caveat to these results is that we use strongly parametrized models, potentially subject to model misspecification. Model misspecification of the $p(q)$ distributions might be particularly important (Fig. 3) as we can resolve the features more precisely in the mass distributions than with spins or redshift. While our results are internally self-consistent when the power-law and normal distributions are swapped, testing with more models and future data can always help shed light on model misspecifications.

However, the strength of the signal and the fact that these transition points have been individually found in other analyses [27,45,46,48,49,124,125] suggests that even with some degree of model misspecification, the presence of three subpopulations and their broad features are real. Furthermore, these results imply that usual assumptions of separability of the population distribution of BBH parameters breaks down above $\sim 25M_\odot$.

Finally, our work focuses mostly on BBHs at higher masses as we do not find any clearly localized subpopulations at $m_1 < 25M_\odot$. In particular, we do not find evidence for any transitions around the $10M_\odot$ peak. There are reasons to expect substructure at this peak, both based on previous observations [14,42,47,126] and based on theoretical considerations. Isolated binary formation channels that undergo stable mass transfer often predict a peak near $10M_\odot$ (for, e.g., Refs. [127–131]). Some studies of cluster formation channels also find a peak around $10M_\odot$ [132,133] while others [134] place the peak at higher masses. We plan to revisit this subpopulation in a future study.

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Data availability—The data and software that support the findings of this article are openly available at [135].

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End Matter

Appendix—

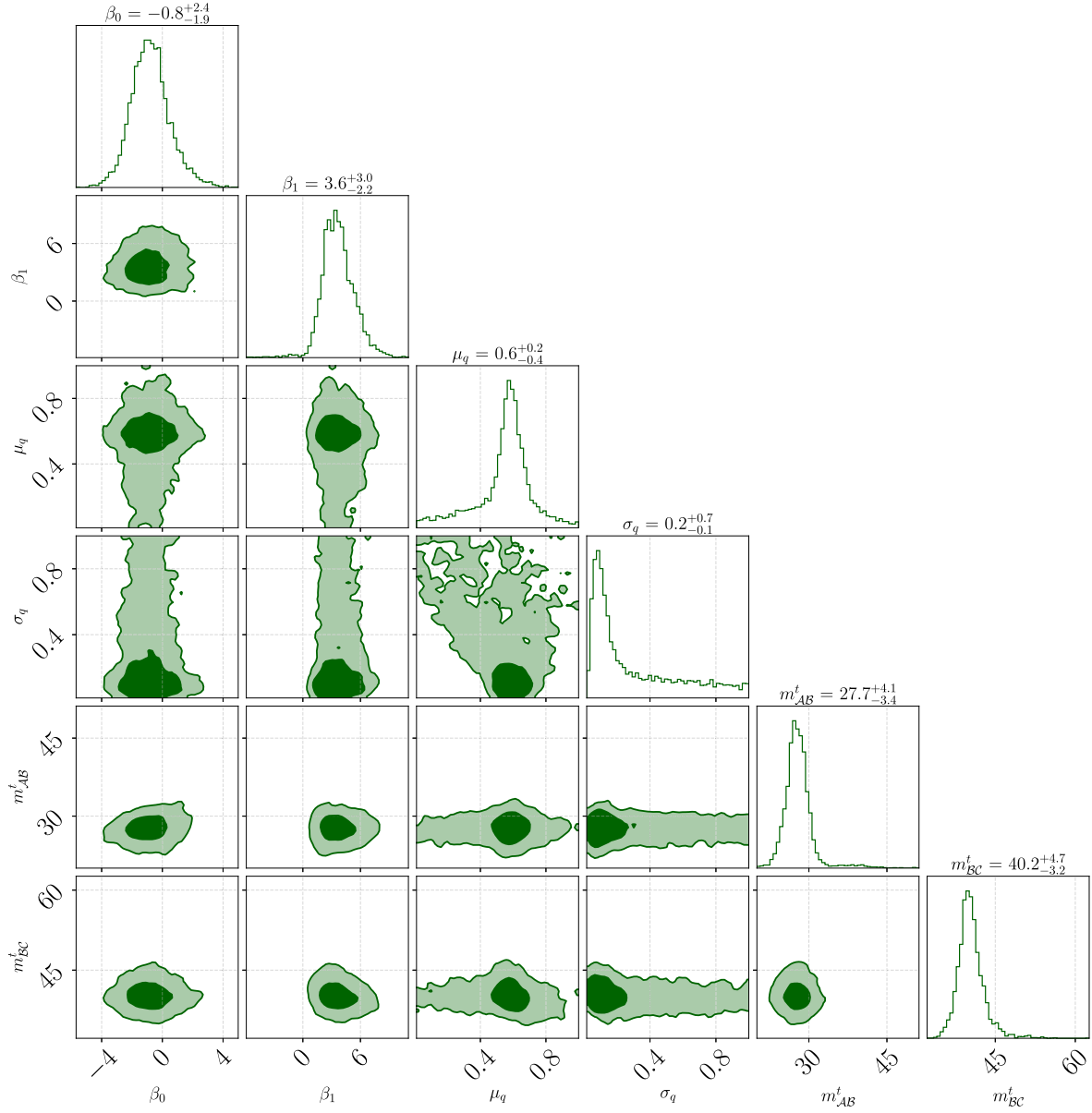


FIG. 5. Corner plot of the mass ratio and transition parameters for the MULTIPOP model.